

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

MARK SCHEME

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

14 September 2015

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	8
2	19
3	16
4	16
5	12
6	9
Section B	
	20
Total	100

This document consists of **16** printed pages.



Section A
Answer **all** questions.

1 Fig. 1.1 shows drawings of a cell at various stages in the mitotic cell cycle.



Fig. 1.1

- (a) List the letters shown in Fig. 1.1 in the order in which these stages occur during a mitotic cell cycle. The first stage has been entered for you.

A **C** **E** **D** **B** [1]

- (b) (i) Describe events occurring in stage D.

1. **centromere divides, kinetochore spindle fibres shorten**

to pull sis chromatids apart to oppo poles of the cell;

2. **non-kinetochore spindle fibres interact to lengthen the cell;**

[2]

- (ii) Outline what happens to the DNA in the nucleus during stage A.

1. **undergoes DNA replication during S phase of interphase;**

2. **DNA molecule unwind & unzip to act as a template for the synthesis of a daughter strand via complementary bp^g;**

3. **semi-conservative replication to result in 2 daughter DNA molecule consisting of 1 parent strand paired with 1 daughter strand;**

4. **joined at centromere to form 2 sis chromatids;**

[3]

- (c) State the importance of mitosis in a multicellular organism, such as a flowering plant or a mammal.

replacement of damaged cells / produce more cells for growth of org;

[1]

- (d) The daughter cell from Fig. 1.1 has 8 chromosomes. State the number of chromosomes a cell from the same organism would have at Prophase I of meiosis.

8;

[1]

[Total: 8]

- 2 During immune response, some B lymphocytes change into plasma cells. Plasma cells are fully differentiated B lymphocytes which produces a single type of antibody.

Fig. 2.1 is a drawing made from an electron micrograph of a plasma cell.

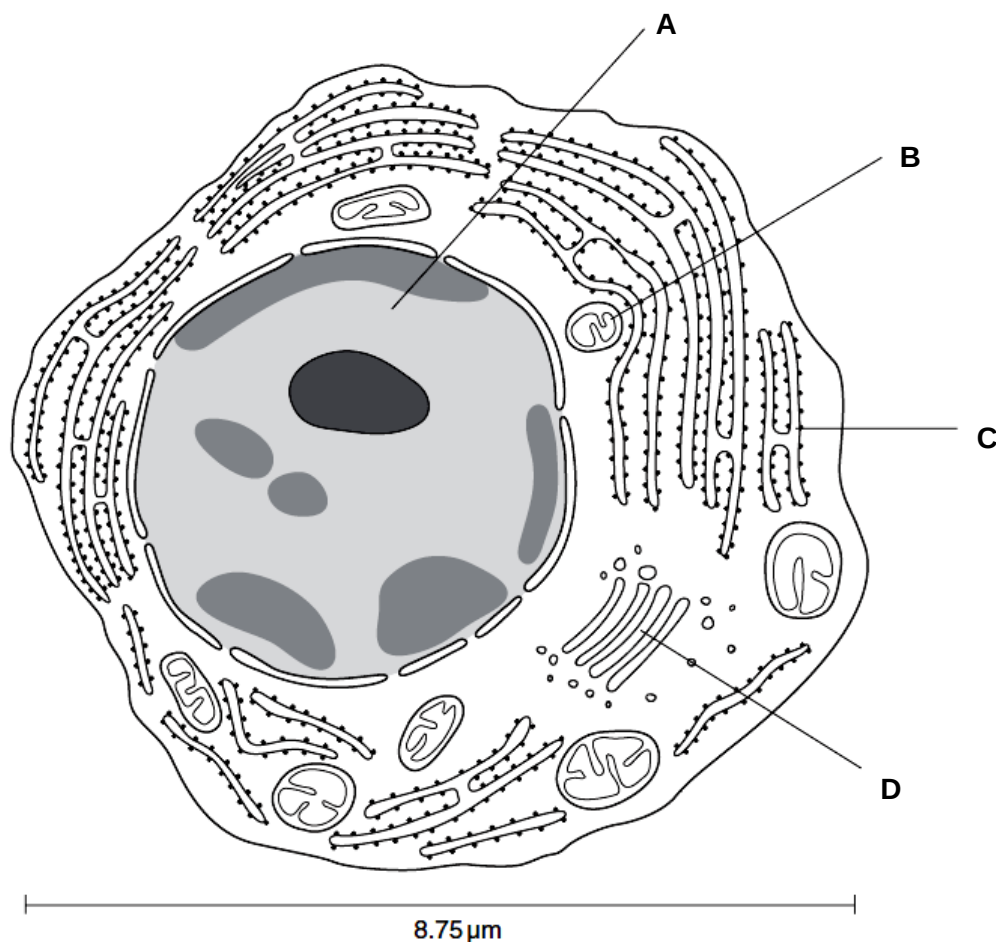


Fig. 2.1

- (a) Calculate the magnification of the drawing in Fig. 2.1. Show your working.

$$\begin{aligned} \text{magnification} &= \text{drawing size} / \text{actual size} \\ 11 \times 10^{-2} \text{ m} / 8.75 \times 10^{-6} \text{ m} \\ &= 12\,571 \text{ times} \end{aligned}$$

[1]

- (b) Identify structures A – D.

A **nucleus;**

mitochondrion;

B **mitochondria**

C **rough endoplasmic reticulum;**

D **Golgi apparatus;**

[2]

- (c) Suggest **one** difference in cellular structures between immature B lymphocyte and a plasma cell.

**more extensive rER / GA; or
more ribosomes;**

[1]

- (d) Describe events leading up to the secretion of antibodies after its translation.

1. **polypeptide enters lumen of rER to fold into its specific 3D config / undergoes glycosylation;**
2. **pinches off ER in trspt vesicle & fuse with cis face of GA;**
3. **Ab is sorted, modified & packaged as it is trspt^d fr cisterna to cisterna;**
4. **where its oligosacc chain has sugar residue added, del or sub to prod a molecular ID tag;**
5. **pinches off at trans face in secretory vesicle;**
6. **translocate to/w CSM of plasma cell with aid of cytoskeleton to fuse with CSM to release AB via exocytosis;**

[4]

Antibodies are proteins produced by B lymphocytes which bind to specific molecules (antigens) on the surface of pathogens like viruses. B cells produce antibodies that are membrane-bound. Each B cell produces a different antibody which recognizes a different antigen.

This antibody, embedded in the cell membrane, protrudes from the B cell acting as a receptor for antigens found on an invading pathogen. During infection, antigens on the pathogen surface bind to the membrane-bound antibody on the B cell thus activating it.

The activated B cell then starts producing antibodies not anchored in the membrane but are secreted into the blood stream where they can aid in fighting infection. Fig. 2.2 shows the different states of B cells.

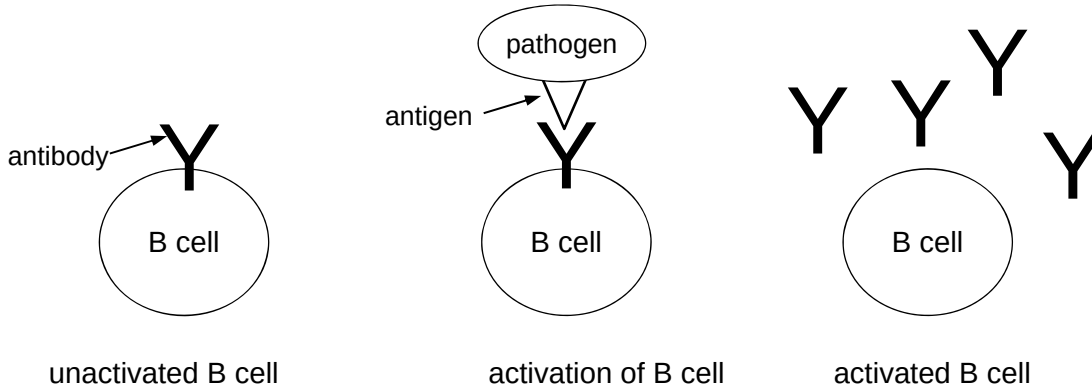


Fig. 2.2

- (e) (ii) Explain how *different antibodies recognize different antigens*.

1. **Ag binding site with specific 3D config**

complementary shape to specific Ag; ® active site

[1]

- (iii) Suggest how antibodies can *aid in fighting infection*.

1. **bind Ag, blocking it from attaching to host cell;**
2. **cause agglutination of virus to allow phagocytes to engulf pathogen;**

or attract / direct other immune cells against pathogen;

[2]

The antibody gene contains 8 exons represented in Fig. 2.3.

Legend: E represents exon
I represents intron

E1	I	E2	I	E3	I	E4	I	E5	I	E6	I	E7	I	E8
----	---	----	---	----	---	----	---	----	---	----	---	----	---	----

Fig. 2.3

The functions of the various exons encoded by the antibody transcript are:

- Exon 1 (E1) encodes a signal peptide which helps target the translated protein into the lumen of the rough endoplasmic reticulum.
- Exon 2 (E2) encodes the variable region which is the part of the antibody that recognizes and binds its specific antigen.
- Exons 3 to 6 (E3, E4, E5, E6) encode the constant region of the antibody. This part of the antibody is the same in different antibodies recognizing different pathogens.
- Exons 7 to 8 (E7, E8) encode a transmembrane domain that anchors the antibody in the B cell membrane.

- (d) Pre-mRNA transcribed from the antibody gene undergoes post-transcriptional modifications to produce a mature mRNA which is translated into antibody. Different forms of antibodies can be produced from the same gene by alternative splicing by spliceosomes.

With reference to Fig. 2.1 and 2.2, illustrate in the space provided

- (i) a pre-mRNA coding for antibody in an **unactivated** B cell;

5' cap	E1	I	E2	I	E3	I	E4	I	E5	I	E6	I	E7	I	E8	polyA
--------	----	---	----	---	----	---	----	---	----	---	----	---	----	---	----	-------

[2]

- (ii) a mature mRNA coding for antibody in an **activated** B cell after post-transcriptional modification.

5' cap	E1	E2	E3	E4	E5	E6	polyA
--------	----	----	----	----	----	----	-------

[2]

- (d) (i) State **two** advantages of alternative splicing in eukaryotes.

1. **same gene locus can code for diff versions of the same prot**

↑ coding capacity of the genome / ↑ diversity of prot encoded by genome;;

2. **able to rapidly prod diff mature mRNAs coding for diff prot from the same pre-mRNA**

more rapid response to Δing env cond / cellular demand;

[2]

(ii) Describe how spliceosome produces a mature mRNA.

1. *snRNA in snRNP binds to splice site at (3' & 5' end) of intron*

via complementary bp^g;

2. *assoc. w other prot to form spliceosome*

that cleaves phosphodiester bonds btw adj nucleotides in intron & exon & ligate exons together;

3. *bind to branch point to form lariat;*

[2]

[Total: 19]

3 Fig. 3.1 shows the arrangement of genes in the lac operon.

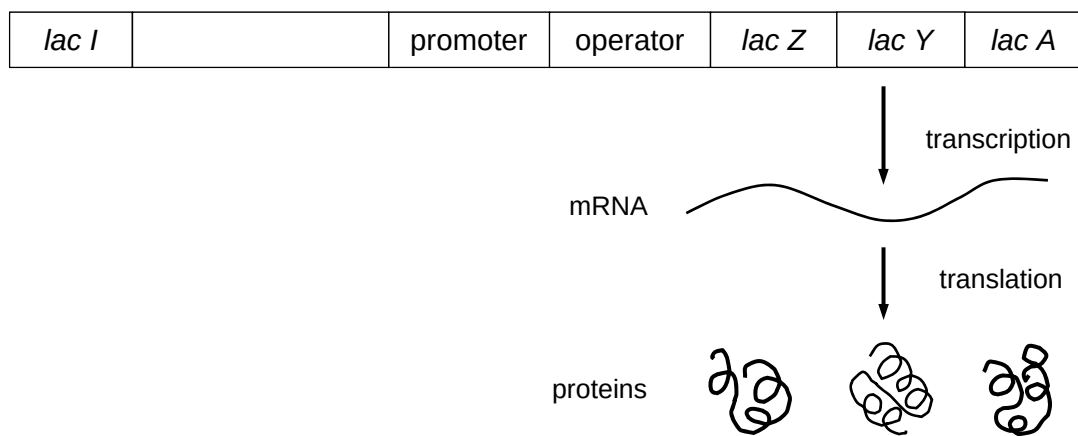


Fig. 3.1

(a) (i) Explain what is meant by an *operon*.

1. *several SG coding for prot/enz governing related metabolic pathway*

under the control of a single promoter & operator;

2. *is switched on & off by binding of repressor prot*

coded by regulatory gene;

[2]

(ii) Name the proteins coded by *lac I*, *Z*, *Y* and *A*.

lac I **repressor prot;**

lac Z **β -galactosidase;**

lac Y **permease;**

lac A **transacetylase;**

[4]

(b) Describe what happens in the presence of glucose and lactose in the nutrient medium.

1. *basal levels of β -gal converts lactose $\rightarrow$ allolac*

which acts as inducer binding to (allosteric site) of (active) repressor;

2. cause conformational Δ in (DNA binding site of) repressor

causing it to detach from operator;

3. *glc* $\oslash$ AC which converts ATP $\rightarrow$ cAMP

resulting in low levels of cAMP-CAP cplx;

4. w/o cAMP-CAP cplx bound at CAP site at promoter;

*RNA pol binds unstably at promoter & $\otimes$ transcribe SG *lac Z, Y, A* or at basal rate;*

[4]

- (c) Explain how a single mRNA produced from transcription of *lac Z, Y* and *A* can be translated into three separate proteins.

1. polycistronic mRNA;

2. with 3 start & stop codes;

[2]

- (d) The *lac Z* gene, together with the *lac* promoter, can be inserted into plasmid vectors used to produce genetically engineered insulin. Explain why the *lac* promoter is inserted instead of insulin promoter.

1. host cell used is prokaryote / bacterium

*RNA pol that binds promoter to transcribe *lac Z* & insulin gene is prokaryotic in origin;*

2. has diff 3D config at DNA binding site

only complementary to DNA seq of bact promoter;

[2]

- (e) Another operon, the arabinose operon, consists of structural genes *ara B*, *ara A* and *ara D* which codes for enzymes that break down arabinose. Explain if arabinose is an inducible or repressible operon.

1. inducible;

2. SGs code for enz involved in catabolic rxns / break down arabinose

thus only switch on when arabinose is present / presence of inducer;

[2]

[Total: 16]

- 4** Glucose-6-phosphate dehydrogenase (G6PD) deficiency is a sex-linked recessive genetic condition. G6PD is involved in the conversion of glucose-6-phosphate to ribose-5-phosphate, producing reduced nicotinamide adenine dinucleotide phosphate (NADPH) in the reaction. NADPH reacts with harmful molecules called reactive oxygen species, protecting cells from damage and destruction.

The production of NADPH by G6PD is especially important in red blood cells, which are particularly susceptible to damage by reactive oxygen species because they lack other

NADPH-producing enzymes. G6PD deficiency causes premature red blood cell lysis leading to haemolysis.

- (a) (i) Explain what is meant by sex-linked disease.

gene locus is found on X chromosome;

[1]

- (ii) Using the information given, explain why only homozygous recessive individuals experience haemolysis.

1. normal G6PD allele is dominant thus expressed regardless of allelic counterpart either in homozygous dominant or heterozygote;

2. 1 copy of normal G6PD allele is enough to code for / prod sufficient G6PD;

3. to convert G6P → ribose-5-phosphate and NADPH which prevents accumulation of reactive oxygen species to cause haemolysis;

4. thus only in homozygous recessive indivs, the mutant G6PD allele codes for non-fnal G6PD leading to haemolysis;

[3]

- (b) Fig. 4.1 is a pedigree showing inheritance of G6PD deficiency in a particular family.

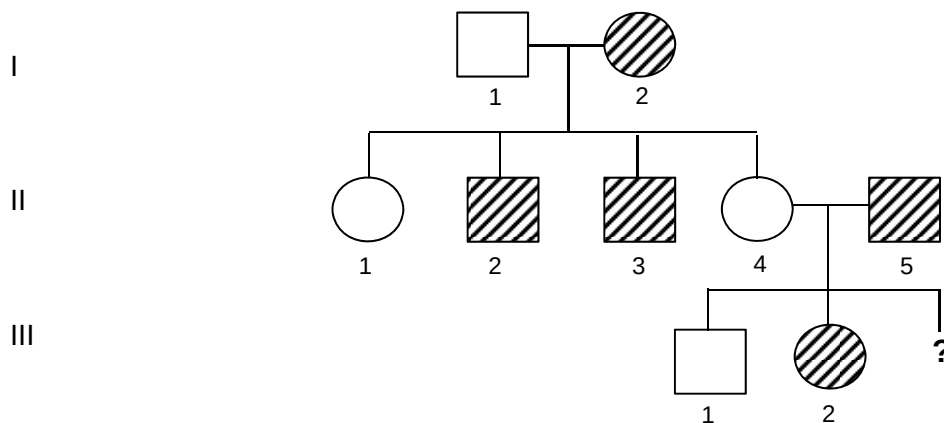


Fig. 4.1

- (b) With reference to Fig. 4.1, cite evidence that the inheritance of G6PD is sex-linked recessive.

1. all sons in gen II of affected mother in gen I are affected as they inherit their only X chromosome maternally;

2. affected mother have X chr w recessive allele which is passed down to sons;

[2]

- (c) (i) Blood group of several individuals are as follows:

I1 blood group A

I2 blood group B

II4 blood group A

II5 blood group O

Using a genetic diagram, illustrate the probability that **II4** and **II5**'s third child would be a daughter with blood group A, who suffers from G6PD deficiency.

Legend

X^G	normal G6DP	I^A	blood group A
X^g	G6PD deficient	I^B	blood group B
		I^O	blood group O

parental genotype ;; $\text{II4 } X^G X^g I^A I^O$ $\times$ $\text{II5 } X^g Y I^O I^O$
gametes;; $X^G I^A$ $X^G I^O$ $X^g I^A$ $X^g I^O$ $X^g I^O$ $Y I^O$

Punnett Sq;;

genotypic & phenotypic ratio;;

state probability = 0.125;;

[5]

- (ii) Explain why the probability of the third child having G6PD deficiency is independent of whether the first and second child suffers from the disease.

1. gametes are produced from separate meiosis where

alleles of G6PD gene undergoes random segregation;

2. each fertilization results from random fertilization;

[2]

- (d) (i) A chi-squared test was conducted to investigate the occurrence of G6PD deficiency in generation II. A χ^2 value of 2.8 was obtained. Using Table 4.1 provided, conclude if the observed ratio deviates significantly from the expected.

Table 4.1

DF	Probability, p				
	0.10	0.05	0.02	0.01	0.01
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82

1. at DF = 1, χ^2 cal of 2.8 is < χ^2 critical of 3.84;

2. $0.05 > p > 0.10$, deviation is statistically insignificant, is due to chance;

[2]

- (ii) Suggest **one** reason why the conclusion in (d)(ii) may not be valid.

small sample size;

[1]

[Total: 16]

- 5 Diabetes mellitus is one of the leading causes of death in Singapore. The International Federation of Diabetes estimates 387 million people worldwide have diabetes, with an increasing trend. There were 533 600 cases of diabetes in Singapore in 2014, with more than 4000 deaths in adults due to the disease.

Diabetes is caused by insufficient or ineffective insulin which functions to lower blood glucose level. Diabetes is a chronic disease with no cure. However, it can be controlled through various means such as a healthy diet, regular exercise, oral medication to balance blood glucose, insulin injections etc.

Insulin, a peptide hormone, is produced by the pancreas when the glucose concentration of the blood is low.

Except in the case of type 2 diabetes, an increase in blood insulin concentration lead to insulin binding to specific sites on the cell surface membrane of target cells. This causes the number of GLUT4 molecules to increase on the cell surface membrane. When blood insulin concentration falls, the GLUT4 molecules are removed from the membrane, making it more impermeable to glucose.

On the onset of type 2 diabetes, the patient's cells become insulin-resistant. Initially the pancreas responds by producing extra insulin. This only partially alleviates the problem of insulin resistance and, in time, overworking of the pancreas leads to their death and subsequently a reduction in insulin production. At this stage the patient may need to receive insulin injections, although this offers only a partial solution.

- (a) Suggest why incidence of diabetes shows an increasing trend over the years worldwide.

1. *changing lifestyles leading to e.g. increase incidence of obesity / any other valid risk factors; or*

2. *high sugar diet leading to freq high bld glc ➡ stress pancreas;*

[1]

- (b) (i) State precisely, where in the pancreas, is insulin produced

β cells in islet of Langerhans;;

[1]

- (ii) State **one** target cell of insulin.

liver cell / muscle cell; ® liver

[1]

- (iii) Identify the *specific sites on the cell surface membrane* that insulin binds to.

(ligand binding site) of receptor tyrosine kinase;

[1]

- (iv) Describe how the number of GLUT4 molecules are increased and removed from the cell surface membrane of target cells.

1. vesicles with GLUT4 embedded in mem translocate to/w CSM

(w aid of cytoskeleton) & fuse w CSM to ↑ no. of GLUT4;

2. infolding / invagination of CSM w GLUT4

to take in GLUT4 by endocytosis;

[2]

(v) Explain why removing GLUT4 molecules would make the cell membrane more impermeable to glucose.

1. glucose is a polar molecule

which is repelled by hydrophobic core of phospholipid bilayer;

2. requires a prot trspter / carrier

to provide a hydrophilic channel;

[2]

Table 5.1 presents the results of an experiment comparing rates of glucose production by a group of people with type 2 diabetes and a control group without the condition, during 23 hours of fasting.

Table 5.1

	rate of glucose production per unit body mass / $\mu\text{mol kg}^{-1} \text{min}^{-1}$		significance level
	patients with type 2 diabetes	control group	
total glucose production	11.1 ± 0.6	8.9 ± 0.5	$p < 0.05$
glucose from hydrolysis of glycogen in the liver	1.3 ± 0.2	2.8 ± 0.7	$p < 0.05$
glucose from gluconeogenesis	9.8 ± 0.7	6.1 ± 0.5	$p < 0.01$

(c) (i) With reference to Table 5.1, explain the difference in glucose production from glycogen and gluconeogenesis in patients with type 2 diabetes and those without.

1. patients w type 2 diabetes produce less glucose fr glycogenolysis than control grp

QV;

2. patients w type 2 diabetes produce more glucose fr gluconeogenesis than control grp

QV;

3. diabetic patients ⊗ prod / respond to insulin to ↑ cellular glucose uptake when bld glucose was high

thus ⊗ have much glycogen store have to prod glucose fr non-carboh sources;

[3]

- (ii) Explain the importance of values of significance level to the experiment.
it indicates the probability that the values obtained for patients & control grp are deviating by chance;

[1]

[Total: 12]

- 6 One laboratory method for investigating possible evolutionary relationships between mammals uses immunology and is summarised in Fig. 6.1.

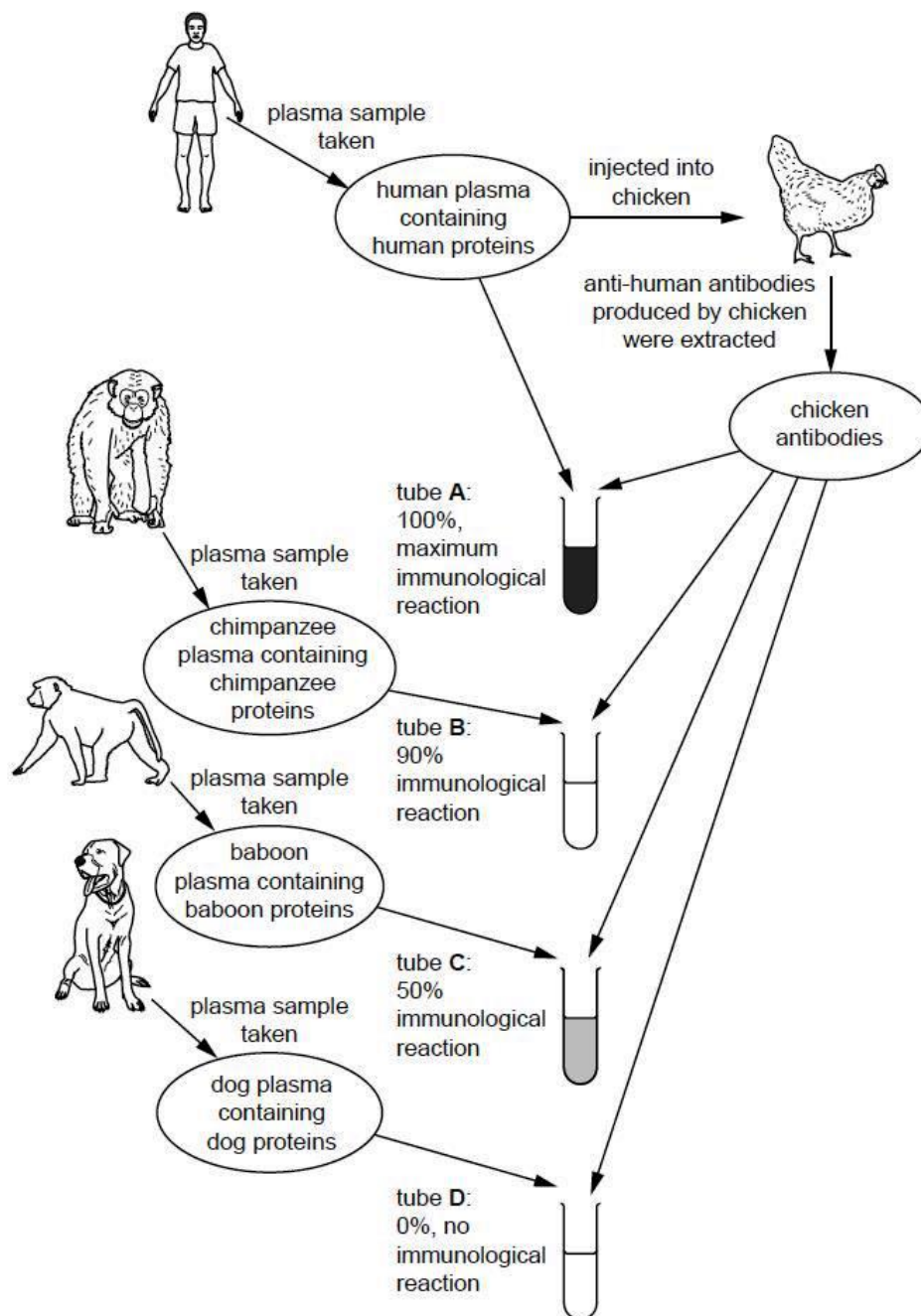


Fig. 6.1

- (a) The immune system of birds is similar to that of mammals.

With reference to Fig. 6.1,

- (i) explain why injecting human proteins into a chicken causes an immune response.

1. **human prot are recognised as foreign / antigens / pathogenic**

by WBCs;

2. **stimulates T/B cells/lymphocytes**

to produce Ab agst it / engulf it;

[2]

- (ii) suggest why investigators described the reaction of the chicken antibodies with the human proteins as 100% (tube A).

1. **tube A act as positive control w the highest % of Ab reacting;**

2. **taken as 100% immunological rxn for comparison with the rest of the samples;**

[2]

- (b) Describe **and** explain the conclusions which could be made about the evolutionary relationships between the **mammals** shown in Fig. 6.1.

[description of evo relationship] (any 2 pt for 1m)

1. **chimpanzees are the most closely related to humans;**

2. **followed by baboon;**

3. **dogs are not (closely) related to humans;**

4. **baboons are closer to human than dogs are to humans;**

[explanation]

7. **chimpanzee and human antigens are (most) similar in terms of aa seq / tertiary struct / 3D shape;**

8. **gives strongest (immunological) response to chicken Ab suggest that human & chimp share recent common ancestor;**

9. **DNA seq / allele coding for Ag is similar w high degree of homology;**

[4]

- (c) State the highest taxon shared in common by organisms shown in Fig. 6.1.

Kingdom; ® Animalia

[1]

[Total: 9]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

- 7 (a) Describe the structure of DNA and how it is packed into chromatin. [8]

Structure (max 4m)

1. **2 polymers of deoxyribonucleotides / polynucleotide chains twisted into a double helix of 2 nm in diameter;;**
2. **held by H bonds btw complementary bases, A=T, C≡G;;**
3. **0.34nm / 10 bp per complete turn;;**
4. **w major & minor groove formed by sugar-phosphate backbone;;**
5. **distinct polarity 3' end with free hydroxyl group at 3C and 5' end with phosphate group at 5C;;**

Packing (max. 4m)

6. **DNA molecule associated 1¾ turns with 8 histone prot / octamer, forming nucleosome core;;**
7. **positively charged histones form ionic bonds with negatively charged DNA (to stabilise struct);;**
8. **nucleosome core associate with H1 histone to form complete nucleosome;;**
9. **linked by spacer/linker DNA forming 10nm 'beads-on-string' structure;;**
10. **further coiled into a solenoid forming a 30nm fibre;;**
11. **further looping in presence of scaffold proteins to form metaphase chromosomes;;**

- (b) Describe what happens to the DNA molecule during S phase of interphase. [7]

1. **DNA replicates via semi-conservative replication to form 2 sister chromatids held at centromere;;**
2. **ea sis chromatid consists of one parent strand paired w one daughter strand;;**
3. **helicase unwinds & unzips DNA double helix at origin of replication (by breaking H bonds btw compl bases);;**
4. **each s/s is bound and stabilized by ssbp at replication fork;;**
5. **topoisomerase prevents supercoiling behind replication fork due to unwinding;;**
6. **primase syn (RNA) primers to provide free 3' OH end to initiate syn;;**
7. **DNA poly III brings in nucleotides w bases complementary to parent / template strand & catalyze formation of phosphodiester bond btw adj nucleotides;;**
8. **elongate daughter strand in 5' → 3' direction by adding nucleoside triphosphates to free 3'-OH end;;**
9. **leading strand is synthesized continuously in the 5' to 3' direction, lagging strand is synthesized discontinuously in Okazaki fragments;;**

10. DNA pol I excise primers & replace with deoxyribonucleotides complementary to parent strand;;

11. Okazaki fragments are joined together by DNA ligase;;

- (c) With a named example, explain how a change in the sequence of the DNA nucleotide may affect the phenotype of the organism. [7]

1. e.g. β -globin gene in Hb;;

2. single base sub of T (in HbA allele) to A (in HbS allele) at the 6th codon;;

3. glutamate w ionic R grp being replaced by valine with a hydrophobic R grp;;

4. results in a hydrophobic protrusion in the β -globin chain;;

5. at low P_{O_2} , β -globin chain forms a hydrophobic pocket which fits the hydrophobic protrusion of a neighbouring Hb molecule;;

6. resulting in aggregation / polymerisation of Hb, leading to crystallization into rod-like fibres;;

7. cause RBC to become sickle-shaped;;

8. sickle-shaped RBC are less elastic thus blocking blood capillaries & interferes with blood circulation to organs;;

9. shorter lifespan of RBC results in higher turnover thus liver damage;;

[Total: 20]

- 8 (a) Carl Linnaeus introduced, in 1735, a system of naming and classifying living organisms just over one hundred years before Darwin published his theory of evolution.

He wrote:

- 'All the species recognized by Scientists came forth from God's hand, and the number of these is now and always will be exactly the same.'
- 'The species are as numerous as the different forms which exist upon this globe – they produce others similar to themselves but in greater numbers.'
- 'Organisms are divided into two kingdoms of nature: viz. the vegetable and animal kingdoms.'
- 'The species and the genus are always the work of nature; the class and the order are the work of nature and art.'

Compare Linnaeus' views of the nature of a species and their classification with those of modern biologists, and account for any differences. [6]

	Linnaeus	modern biologist
1. origin	creationist context / created by God	evolved from existing forms
2. no. of spp	no. of species remain the same	species undergo extinction & new species are formed by diff speciation modes
3. genetic basis	descendants are the same as ancestors	populations show descend with modification
4. descendants	produce offspring in greater numbers	no. of offspring is dependent on survival rate & repro success due to natural selection

5. <i>kingdoms</i>	<i>only 2 kingdoms: vegetable & animal</i>	<i>5 kingdoms: Animalia, plantae, fungi, prokaryote, protist</i>
6. <i>taxonomy</i>	<i>grp organisms into taxa of class, order, family, genus, species by characteristics</i>	<i>grp organisms into taxa taking into account phylogeny and common ancestry / evolutionary history</i>

- (b) Comment on the statement 'Genetically modified crop plants and animals should be treated as new species'. [6]

- 1. GMOs are organisms that have acquired, by artificial means 1/> genes fr an/or organism usually from diff spp;*
- 2. transgenes code for modified or new / novel gene products giving new phenotypic trait / characteristics;*
- 3. BSC ➡ popn of organisms that have the potential to interbreed to produce viable, fertile offspring;*
- 4. GMOs with transgenes that ⊗ affect ability to interbreed with wild type spp e.g. Bt toxin producing crop plants would not be considered a new spp;*
- 5. However some GMOs e.g. GM salmon which ⊗ interbreed with wild salmon ➡ ⊗ be classified as new spp under BSC as they are GM to be triploid thus sterile;*
- 6. MSC ➡ popn of organisms are similar in morphology / physical / anatomical structures;*
- 7. GMOs with transgenes mostly ⊗ confer significantly novel / diff morphological struct / anatomical Δes;*
- 8. e.g. Golden rice has diff colour but basically similar endosperm, embryo struct;*
- 9. e.g. GM salmon is just diff in size, a characteristic that varies in natural popns;*
- 10. ESC ➡ popn of organisms that occupy the same ecological niche;*
- 11. Transgenes may allow the GMO to occupy a different ecological niche e.g. GM salmon with AFP which allows it to survive in colder waters OR glyphosate resistant crop able to survive in presence of herbicide ➡ new spp;*
- 12. Transgenes ⊗ allow GMO to occupy new ecological niche e.g. Flavr Savr tomato with delayed ripening times ➡ not new spp. ;*
- 13. PSC ➡ smallest / irreducible grp of organisms sharing a common ancestor;*
- 14. GMOs still share the wild type as the same common ancestor after intro of transgene;*
- 15. still share anatomical, embryological & molecular homology as wild type ➡ not new spp;*

- (c) Explain how homology supports Darwin's theory of natural selection. [8]

- 1. inherited from a common ancestor thus has same basic struct;;*
- 2. but play diff fn as a result of natural selection to adapt to diff habitats;;*
- 3. anatomical homology e.g. pentadactyl limb with same 5 sets of bones but form wings for flight in bats, fins for swimming in whales;;*
- 4. vestigial organs e.g. post-anal tail present in monkeys for balance but reduced to tail bone in humans;;*

5. *embryological homology e.g. pharyngeal gill slits form for all organism during embryological devt but close up in terrestrial organisms;;*
6. *the more closely related organisms are, the longer the embryological development remains similar;;*
7. *molecular homology e.g. closely related spp with fewer diff in base / aa seq of cytochrome c / haemoglobin gene / prot;;*
8. *due to more recent divergence thus less time to accumulate diff mutations;;*

[Total: 20]